

PlantAL



An Interactive Active Learning System
for Adaptive Tomato Disease Diagnosis

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Interactive Machine Learning · 2025–2026

Active Learning

Transfer Learning

LLM Feedback

Browser-based

Introduction



The Problem

Tomato diseases cause major crop losses. Farmers need fast, accurate diagnosis, but expert agronomists are rarely available in rural areas.



Dataset Gap

Models trained on PlantVillage fail in real fields: different lighting, camera quality, and regional disease strains make them unreliable.



IML Solution

Interactive Machine Learning places the human expert in the loop. The model learns from targeted feedback, not just bulk labeled data.

Scenario & Users



Agricultural Extension Officer

Medium technical · High domain knowledge

Goal: Build a region-specific disease classifier without ML expertise

Task: Labels the most uncertain images first, actively steers the model



Smallholder Farmer

No technical knowledge · High local knowledge

Goal: Get a diagnosis and treatment plan for diseased plants

Task: Reports treatment outcome in plain language, feedback trains the system

Application Design



Tab 1: Label & Train

- 1 Upload batch of unlabeled field photos
- 2 Features extracted instantly (MobileNet)
- 3 Most uncertain image shown first
- 4 User labels → auto-retrain every 5 labels
- 5 Entire pool rescored & reranked after retrain



Tab 2: Field Assistant

- 1 Farmer uploads single field photo
- 2 Model diagnoses disease + confidence
- 3 LLM generates personalized treatment
- 4 Farmer applies treatment, writes/chooses feedback
- 5 Feedback stored → LLM adapts next time

Implementation

Classifier

MobileNetV1 (0.25×) + trainable
Dense head

Training

[TensorFlow.js](https://www.tensorflow.org/js), runs entirely in
browser, no server

Dataset

60 images, 10/class, 6 tomato
disease categories

Treatment v1

Bayesian Bandit (Thompson
Sampling), offline

Treatment v2

Groq API / Llama 3, LLM with
feedback context.

Field Assistant — live demo

● Early Blight

92%

ABOUT THIS DISEASE

Caused by *Alternaria solani*. Dark brown spots with concentric rings on older leaves. Can reduce yield by 50% if untreated.

RECOMMENDED TREATMENT

1. Apply mancozeb as broad-spectrum alternative

67% · 1 report

2. Use azoxystrobin systemic fungicide

67% · 1 report

3. Remove and destroy infected leaves

50% · 4 reports

4. Apply copper-based fungicide as alternative

50% · 4 reports

5. Apply chlorothalonil fungicide immediately

50% · 4 reports

Pilot Results

60

Images in pool

62

Labels given

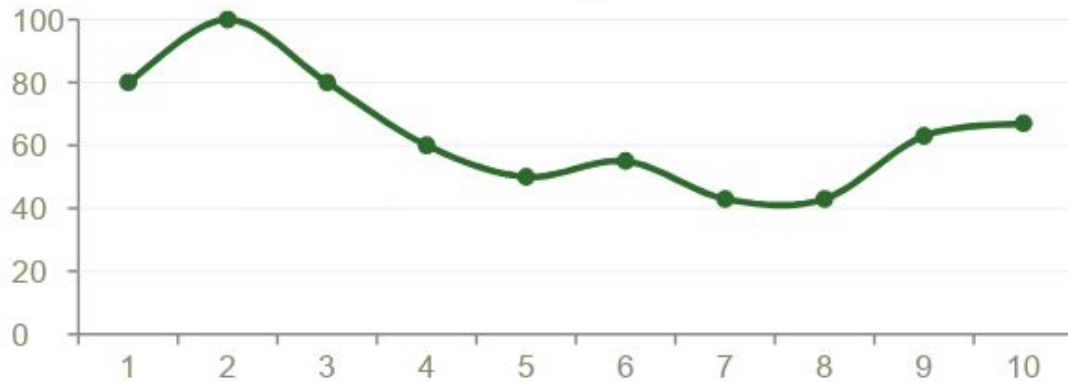
10

Retrains

67%

Final val acc.

Validation Accuracy per Retrain



Key Observations

- Acc. climbs: 43% → 67% (R8→R10)
- Val split activated at 20+ samples
- Fine-tuning kept weights across retrains
- Class imbalance (9× Yellow Curl) caused plateau
- Coverage constraint needed in production

Evaluation Plan

Method:

Within-subjects comparison, same participants complete two labeling sessions: one with uncertainty sampling (PlantAL) and one with random ordering. Participants are agricultural extension officers from a regional cooperative.



Label Efficiency

Accuracy vs. number of labels, learning curve comparison



Time to Threshold

Labels needed to reach 80% val accuracy



User Workload

NASA-TLX questionnaire after each session



Treatment Adoption

% of LLM recommendations farmers actually tried

H1: Uncertainty sampling reaches 80% accuracy with fewer labels than random.

Conclusion

- Full active learning loop running in the browser, no server needed
- Two treatment modules: offline bandit + LLM with free-text farmer feedback
- Multi-model pipeline: MobileNet (vision) + Llama 3 (language)
- Pilot: 67% val accuracy on 6 classes with 62 labels
- Future: persistent storage, class coverage constraints, formal evaluation

Thank you

